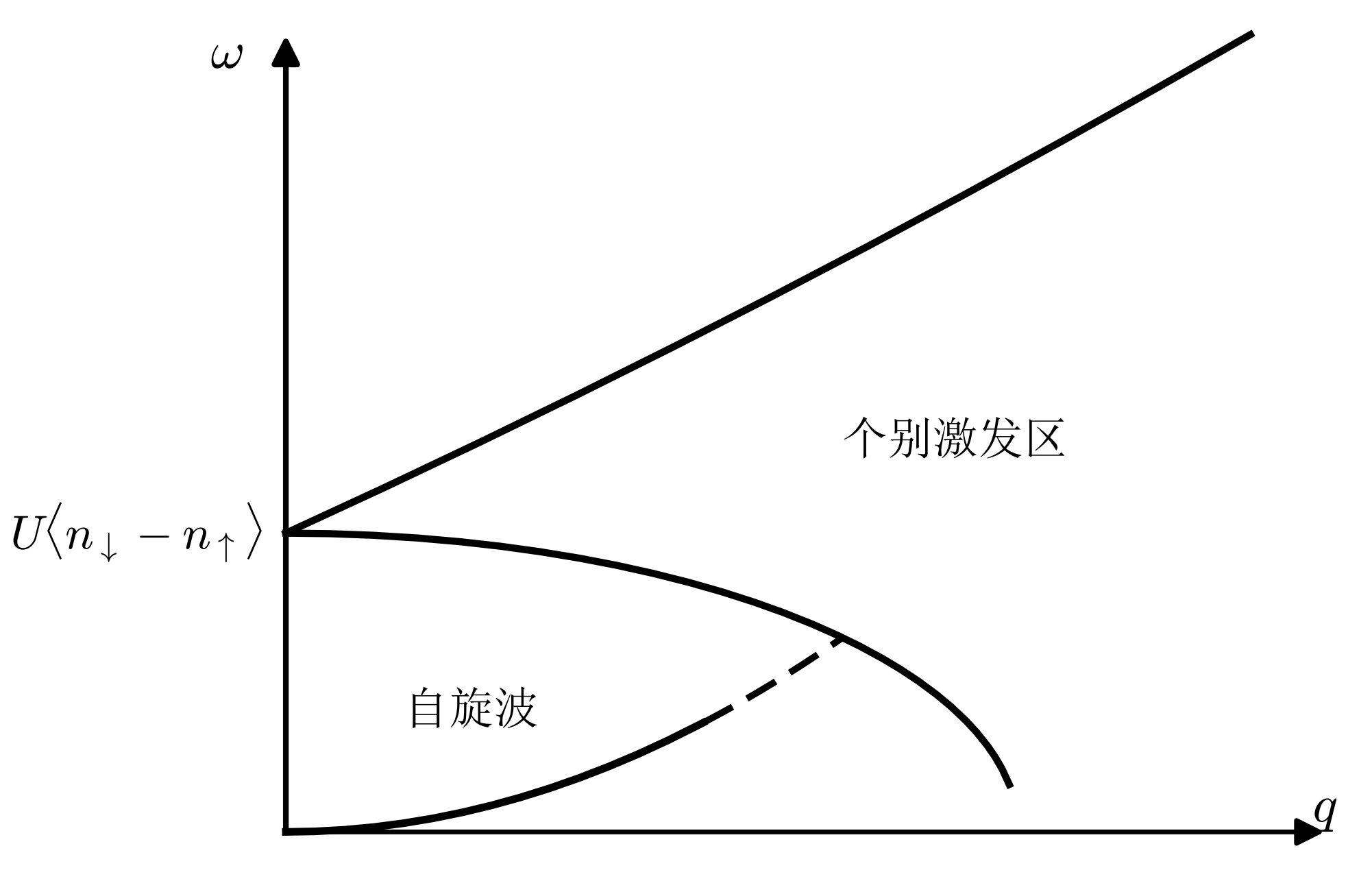
ω
$U\langle n_\downarrow - n_\uparrow \rangle$
个别激发区
自旋波
q